

Plant cell

The major difference between the cells of plants and animals is that plant cells are surrounded by a cell wall made of a lattice of cellulose strands. The space between the walls of neighboring cells is called the middle lamella. It contains a cement made of pectin, a sugary gel that joins the cells together.

Vesicles

A membrane sac that can store or transport substances.

Golgi apparatus

This bags up substances into vesicles.

Mitochondrion

This creates the cell's power supply.

Cell wall

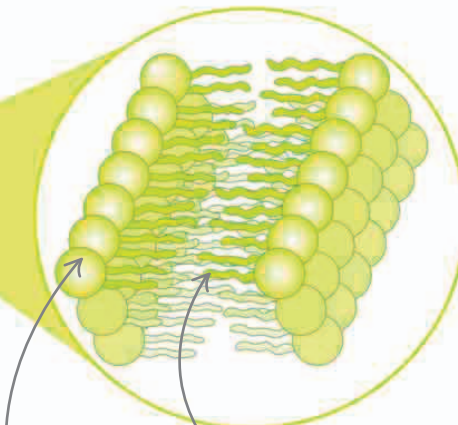
A lattice of cellulose, a tough polymer made from chains of glucose.

Cell membrane

The membrane is not attached to the wall, and moves as the cell shrinks and swells.

▽ Membrane structure

The cell's outer layer, or membrane, is selectively permeable—it allows only some things to enter and leave the cell. The membrane is made from double layers of fat chemicals called lipids. The "head" of a lipid is hydrophilic, meaning it mixes with water and substances on each side of the cell. The "tail" is hydrophobic—it is repelled by water, and forms a barrier that helps keep the cell's contents inside.



Lysosome

A bag of destructive enzymes that break down unwanted materials in the cell.

Hydrophilic head

The heads float in the cytoplasm and extracellular liquids.

Hydrophobic tail

The two lipid layers connect by their tails to form a thin, water-repellent film on either side of the membrane.

Chloroplast

Folded membranes covered in chlorophyll, a green pigment found in plants.

Nucleus

Contains the nucleolus, which makes ribosomes.

Ribosome

The site where proteins are made.

Vacuole

A container for storing water, which also gives the cell structure.

Druse crystal

A crystal of calcium oxalate, which makes plants less palatable to herbivores.

Amyloplast

This turns sugars into starches.

△ Plant cell construction

Plant cells largely contain the same kinds of organelles as animal cells. The main additions are the chloroplasts in the cells of green sections of the plant body. This is where photosynthesis occurs, the process that produces the plant's sugar fuel.

REAL WORLD

Microscopic cells

Most cells are not visible to the naked eye, so microbiologists study them through microscopes. The first person to see cells in this way was 17th-century English scientist Robert Hooke. He named them cells after the small rooms used by monks. Today, microbiologists use dyes and lighting techniques to show a cell's internal structure, such as these human body cells (below).

